University of Central Punjab
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FACULTY OF INFORMATION TECHNOLOGY

Quiz 2

CLO	CLO Statement	Learning Domain	PLO
2	Apply software design principles, refactoring techniques and design patterns to develop	C3	P4

Questions:

1. Explain the difference between "bad code" and "clean code" with respect to naming conventions, using an example from your experience or the provided material.

2. Define what is meant by "code smells" and list at least three common types of code smells that developers should look for when refactoring Java code.

3. Explain the importance of following the Single Responsibility Principle (SRP) when designing classes in Java. How does SRP contribute to maintaining clean and maintainable code?

4. Explain the use of the constant in the following Java code. How does replacing a magic number with a constant improve code readability and maintainability?

```
public class CircleArea {
public static void main(String[] args) {
double radius = 5;
double area = 3.14 * radius * radius;
System.out.println("Area: " + area);
}
}
```

5. Explain how the Dependency Inversion Principle (DIP) can be applied to a scenario where a high-level module (such as a Switch class) directly depends on a low-level module (such as a LightBulb class). Refactor the code to show how DIP can reduce tight coupling between these modules.